Customer Shopping Behavior Analysis

Data Analysis Report · Python · PostgreSQL · Power BI

1. Executive Summary

This report analyzes 3,900 customer purchase records to understand spending patterns, customer segments, product preferences and subscription behavior. The data was cleaned in Python, analyzed with ten business queries in PostgreSQL, and summarized in an interactive Power BI dashboard.

Headline findings

- **Total revenue is $233,081**, with an average purchase of **$59.76** and an average review rating of **3.75 / 5**.
- **Clothing dominates**: it accounts for 1,737 of 3,900 purchases (44.5%) and is the largest revenue category, followed by Accessories, Footwear and Outerwear.
- **Male customers generate 67.7% of revenue** ($157,890 vs. $75,191), but this is driven by customer volume (68% of customers are male), not by higher spend per purchase.
- **Subscribers do not spend more**: average spend is $59.49 for subscribers vs. $59.87 for non-subscribers. Only 27% of customers subscribe.
- **Repeat buyers are not more likely to subscribe**: 27.6% of customers with more than five previous purchases are subscribers, essentially the same as the 27.0% overall rate.
- **43% of purchases used a discount**, and Hat, Sneakers and Coat are the most discount-dependent products.
- **Revenue is evenly spread across age groups** (23.9% to 26.7% each), with Young Adults slightly ahead.

2. Objectives and Scope

The analysis set out to answer the following business questions:

- Who are our customers, and which segments contribute the most revenue?
- Which products and categories drive sales and satisfaction?
- How much do discounts, shipping type and subscriptions influence spending?
- Where are the opportunities to increase loyalty and subscription uptake?

3. Data Overview

Attribute	Detail
Records	3,900 purchases (one row per customer ID)
Columns	18
Demographics	Age, Gender, Location, Subscription Status
Purchase details	Item Purchased, Category, Purchase Amount (USD), Season, Size, Color
Behavior	Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type, Payment Method
Missing data	37 values in Review Rating (0.9%)

Summary statistics (selected)

Metric	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
Mean	44.07	59.76	3.75	25.35
Std. deviation	15.21	23.69	0.72	14.45
Min	18	20	2.5	1
Median	44	60	3.8	25
Max	70	100	5.0	50

Other profile facts: 2,652 customers are male (68%), there are 4 product categories, 25 distinct items, 50 locations, and 6 shipping types. The most common item is Blouse (171 purchases), the most common payment method is PayPal (677), and the most common purchase frequency is Every 3 Months (584).

4. Methodology

4.1 Data preparation (Python / pandas)

1. **Loading and inspection**: loaded the dataset with pandas and reviewed structure and distributions with `df.info()` and `df.describe()`.

2. **Missing values**: the 37 missing Review Rating values were imputed with the **median rating of each product category**, which keeps the imputation sensitive to category differences and robust to outliers.
3. **Standardization**: column names were converted to snake case (e.g. `purchase_amount`, `subscription_status`).
4. **Feature engineering**: created `age_group` (binned ages) and `purchase_frequency_days` (numeric form of purchase frequency).
5. **Consistency check**: `discount_applied` and `promo_code_used` were found to be redundant, so `promo_code_used` was dropped.
6. **Database load**: the cleaned DataFrame was loaded into PostgreSQL for SQL analysis.

4.2 Analysis (PostgreSQL)

Ten business questions were answered with SQL using aggregation, conditional logic (`CASE`), subqueries and window functions (ranking within categories).

4.3 Visualization (Power BI)

An interactive dashboard with KPI cards, subscription, category and age-group views, and slicers for subscription status, gender, category and shipping type.

5. Findings

5.1 Revenue by gender

Gender	Revenue (USD)	Share
Male	157,890	67.7%
Female	75,191	32.3%

Males contribute roughly twice the revenue of females. However, with 2,652 male and 1,248 female customers, average revenue per customer is nearly identical (about $59.54 male vs. $60.25 female). The gap reflects **who is buying, not how much each buyer spends**, which points to an acquisition opportunity among female customers rather than a spending problem.

5.2 Revenue by age group

Age group	Revenue (USD)	Share
Young Adult	62,143	26.7%
Middle-aged	59,197	25.4%
Adult	55,978	24.0%
Senior	55,763	23.9%

The spread between the highest and lowest group is only about 11%. No single age group dominates, so age alone is a weak targeting variable.

5.3 Subscribers vs. non-subscribers

Status	Customers	Avg. spend (USD)	Total revenue (USD)
Subscribed	1,053 (27%)	59.49	62,645
Not subscribed	2,847 (73%)	59.87	170,436

Subscribers spend no more per purchase than non-subscribers, and 73% of customers are not subscribed. The subscription program currently shows no spending uplift.

5.4 Repeat buyers and subscriptions

Among 3,476 customers with more than five previous purchases, 958 (27.6%) are subscribers and 2,518 are not. This is virtually the same as the overall subscription rate (27.0%), so **purchase history does not predict subscription**. Thousands of engaged repeat buyers are an untapped conversion pool.

5.5 Customer segmentation

Segment	Customers	Share
Loyal	3,116	79.9%
Returning	701	18.0%
New	83	2.1%

About 80% of customers fall in the Loyal segment and only 2% are New. This suggests either a mature customer base with weak new-customer acquisition, or segment thresholds that are too generous to differentiate customers. The thresholds should be reviewed (see Limitations).

5.6 Shipping type

Shipping type	Avg. purchase (USD)
Express	60.48
Standard	58.46

Express shipping users spend about $2.02 (3.5%) more per purchase, a modest but consistent premium.

5.7 Products and categories

Top 5 products by average rating

Rank	Item	Avg. rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Most discount-dependent products (share of purchases with a discount)

Rank	Item	Discount rate
1	Hat	50.00%
2	Sneakers	49.66%
3	Coat	49.07%
4	Sweater	48.17%
5	Pants	47.37%

Top products by category (by orders)

Category	#1	#2	#3
Accessories	Jewelry (171)	Sunglasses (161)	Belt (161)
Clothing	Blouse (171)	Pants (171)	Shirt (169)
Footwear	Sandals (160)	Shoes (150)	Sneakers (145)
Outerwear	Jacket (163)	Coat (161)	n/a

Top ratings are tightly clustered (3.78 to 3.86), so differences between products are small. Hat is notable for appearing in both the top-rated list and as the most discount-dependent item, suggesting it is well liked but may be under-priced. Discount dependence is high across the board (47% to 50% for the top five).

5.8 High-spending discount users

839 customers used a discount and still spent above the average purchase amount. These customers respond to promotions without sacrificing basket size, making them a good group for targeted, controlled offers.

5.9 Dashboard

The Power BI dashboard presents 3.9K customers, a $59.76 average purchase and a 3.75 average rating, alongside subscription split (27% / 73%), revenue and sales by category, and revenue and sales by age group, with filters for subscription, gender, category and shipping type.



Figure 1. Customer Behavior Dashboard (Power BI).

6. Recommendations

Priority	Recommendation	Evidence	Suggested KPI
High	**Rework the subscription value proposition** with benefits that justify membership (free or express shipping, early access, member pricing)	Subscribers spend the same as non-subscribers; only 27% subscribe	Subscription rate; subscriber average spend
High	**Convert repeat buyers to subscribers** through targeted campaigns	2,518 repeat buyers are not subscribed	Conversion rate of repeat buyers
High	**Review discount policy**, especially for Hat, Sneakers, Coat, Sweater and Pants	About half of purchases of these items are discounted	Margin per item; discount rate
Medium	**Grow the female customer base** with targeted acquisition	Female spend per customer matches male, but female customers are 32% of the base	Share of female customers

Priority	Recommendation	Evidence	Suggested KPI
Medium	**Redefine loyalty tiers** and run a loyalty program with meaningful differentiation	80% of customers are labeled Loyal; only 2% are New	Segment distribution; repeat rate
Medium	**Promote Express shipping** as an upsell	Express buyers spend 3.5% more	Express adoption; average order value
Low	**Feature top-rated and best-selling items** in campaigns	Gloves, Sandals, Boots, Hat, Skirt; Jewelry, Blouse, Jacket lead their categories	Campaign conversion
Low	**Use category rather than age for targeting**	Age groups contribute near-equal revenue; Clothing leads categories	Revenue by category

7. Limitations and Next Steps

- **No time dimension**: the data has no purchase date, so trends, seasonality (beyond the Season field) and cohort analysis are not possible.
- **No cost or margin data**: discount recommendations rely on discount frequency, not profitability.
- **One row per customer**: "customers" and "purchases" are effectively the same unit here, so lifetime value cannot be measured directly.
- **Segment thresholds**: the cut-offs for New, Returning and Loyal are not documented in the source and produce a very skewed split.
- **Imputation**: median imputation for 37 ratings (0.9%) is low-risk but still an assumption.
- **Differences are small**: gaps such as Express vs. Standard or between top-rated products are modest, and statistical tests (t-tests, chi-square) should confirm significance before acting on them.

Suggested next steps: add statistical significance testing, build a subscription propensity model (e.g. logistic regression), run a customer clustering (RFM or k-means), and include cost data to quantify discount impact on margin.

8. Appendix: Tools

Stage	Tool
Cleaning and feature engineering	Python (pandas)
Storage and business queries	PostgreSQL
Dashboard	Power BI